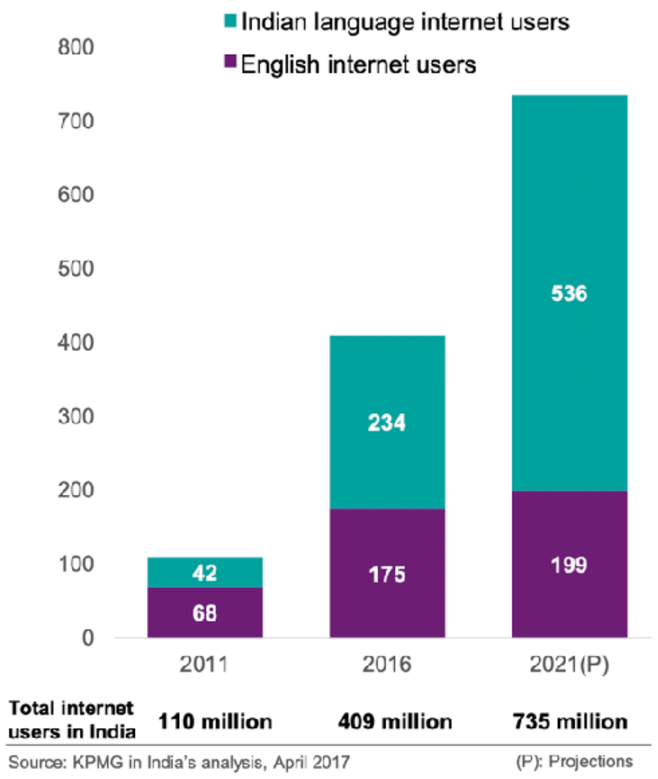
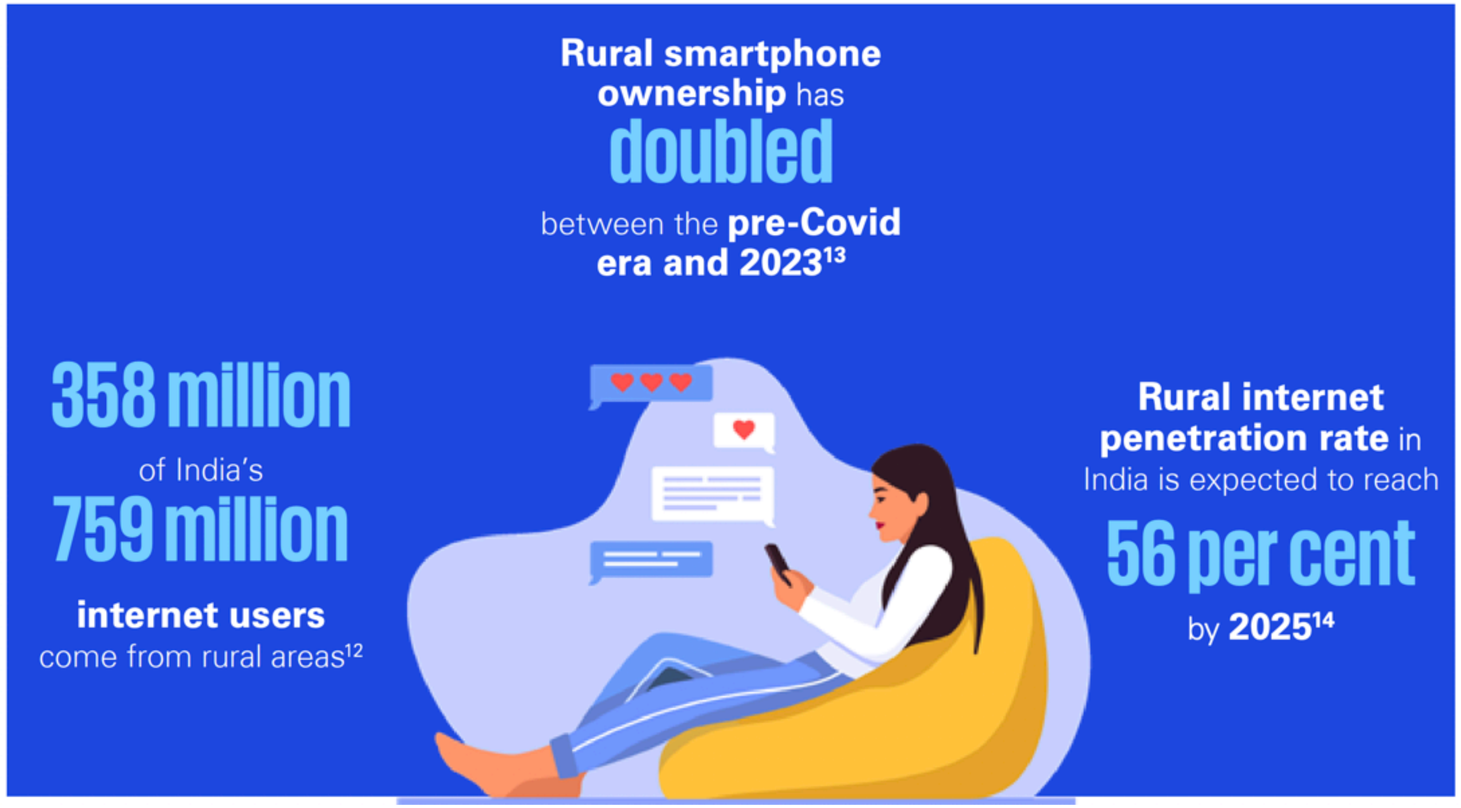
Speech and Multilinguality in Large Language Models

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April 10, 2026

Growing Importance of Multilingual Speech and Language Technologies

- Internet users in India largely prefer their primary language over English (***multilinguality***)





- User interactions are largely over smartphones (***speech***)
- Critical to improve LLM support for both spoken language and multilingual use

Image from "A study by KPMG in India and Google", April 2017

Image from "Expanding tech horizons: Unlocking India's digital potential", October 2023

Multilinguality in LLMs

Multilingual LLMs

- Multilingual LLMs (**MLLMs**): Query and generate in multiple languages

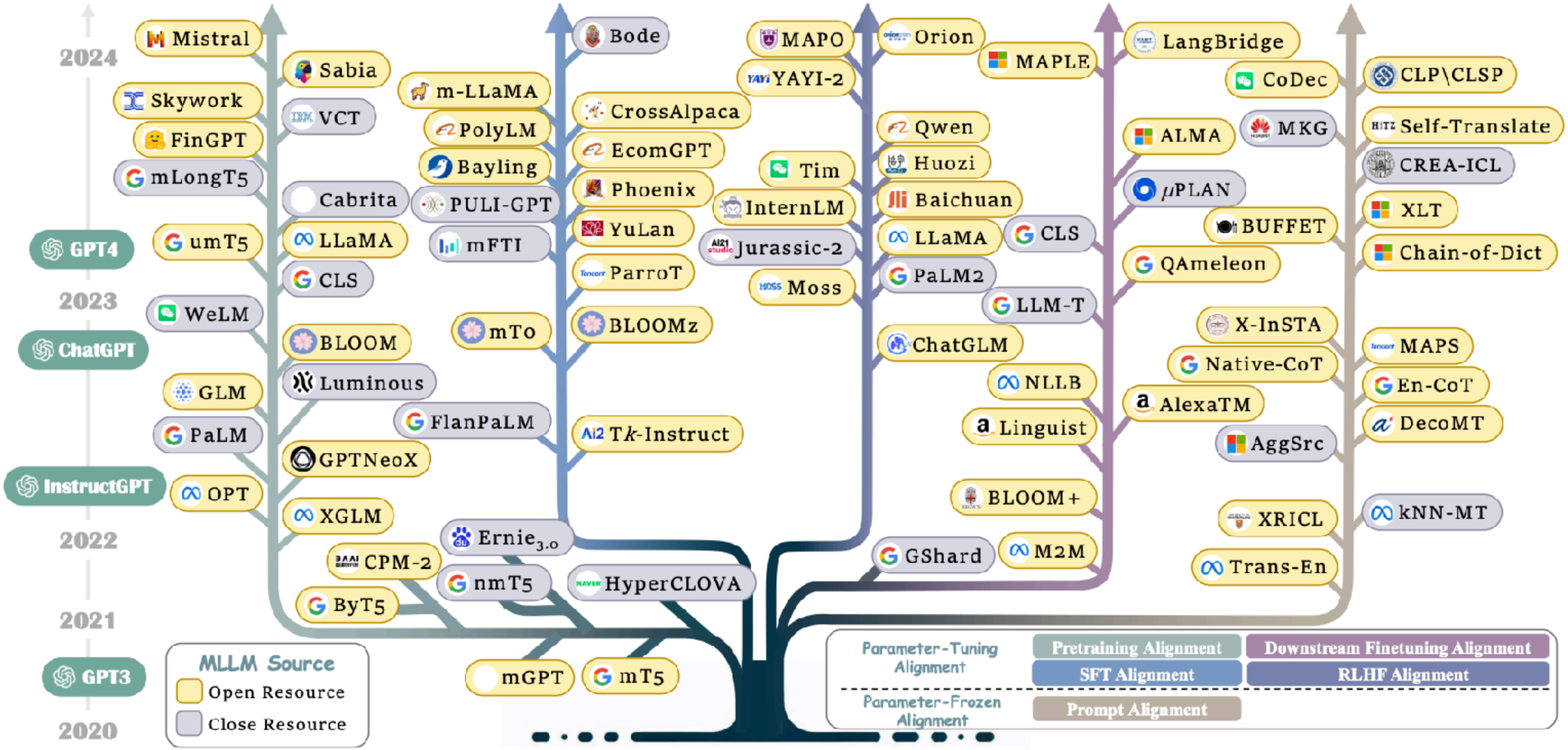


Image from "Multilingual Large Language Model: A Survey of Resources, Taxonomy and Frontiers", Qin et al., arxiv, April 2024

MLLMs Lag Behind on (Most) Languages other than English

- Significant performance gaps between English and other languages (esp. low-resource covering nearly all Indian languages) on standard NLP tasks

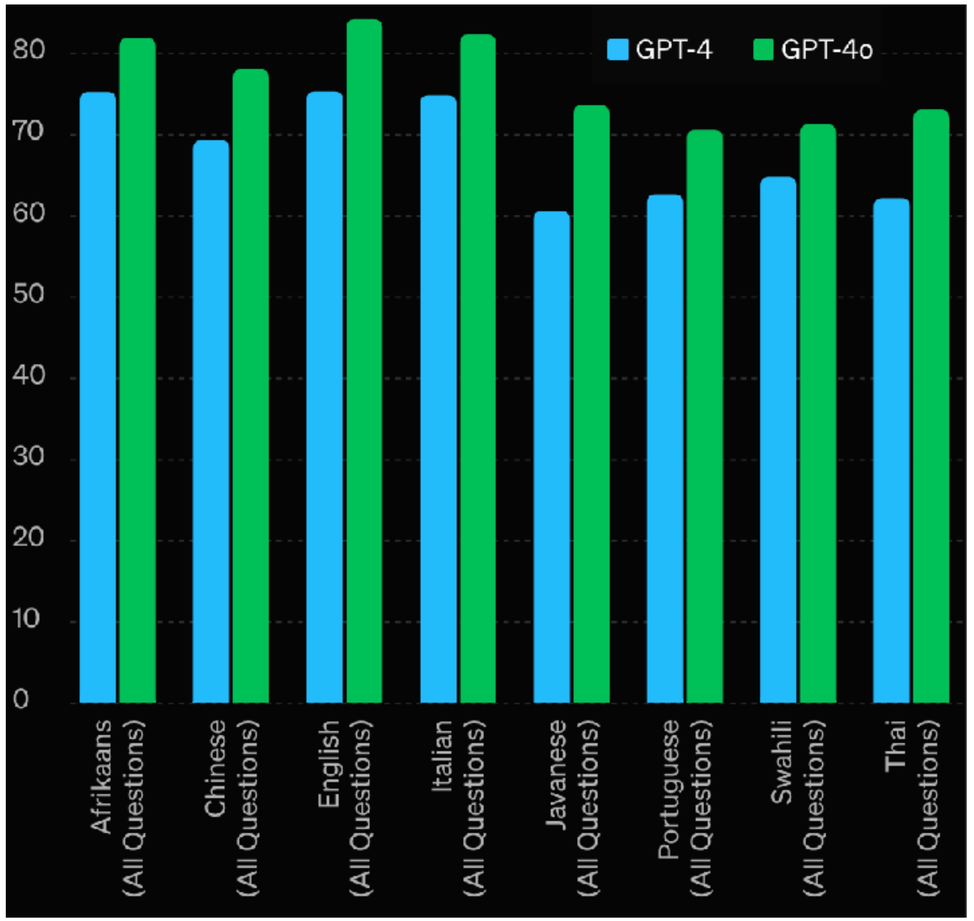
XQuAD

Language	ChatGPT(en)	
	EM	F1
English	56.0	74.9
Russian	30.2	49.1
German	45.9	65.8
Chinese	37.1	42.3
Spanish	41.8	65.8
Vietnamese	36.1	57.3
Turkish	34.5	56.4
Arabic	32.0	50.3
Greek	29.7	45.0
Thai	31.2	43.4
Hindi	17.5	37.8

XNLI

Language	ChatGPT	
	(en)	(spc)
English	70.2	70.2
Russian	60.8	45.4
German	64.5	51.1
Chinese	58.2	35.5
French	64.8	42.2
Spanish	65.8	47.4
Vietnamese	55.4	44.8
Turkish	57.1	37.1
Arabic	55.3	22.3
Greek	55.9	54.5
Thai	44.7	11.5
Bulgarian	59.7	44.6
Hindi	48.8	5.6
Urdu	43.7	6.3

M3Exam

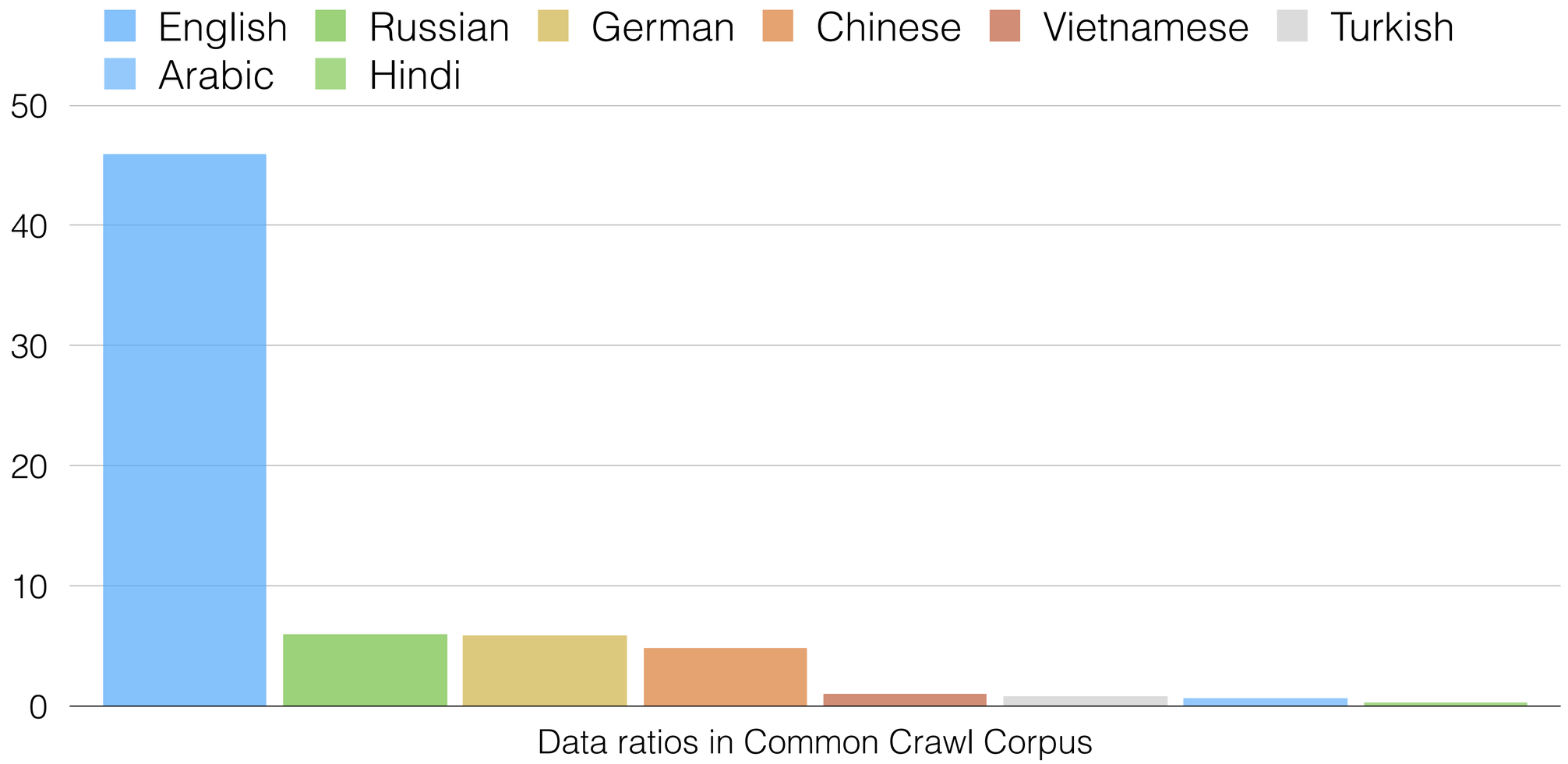


XQuAD/XNLI results: "ChatGPT Beyond English: Towards a Comprehensive Evaluation of Large Language Models in Multilingual Learning", Lai et al., EMNLP 2023

GPT-4o: https://openai.com/index/hello-gpt-4o/

MLLMs Lag Behind on (Most) Languages other than English

- Primary causes include low diversity in pretraining data



MLLMs Lag Behind on (Most) Languages other than English

- Primary causes include low diversity in pretraining data, tokenizer fertility, etc.

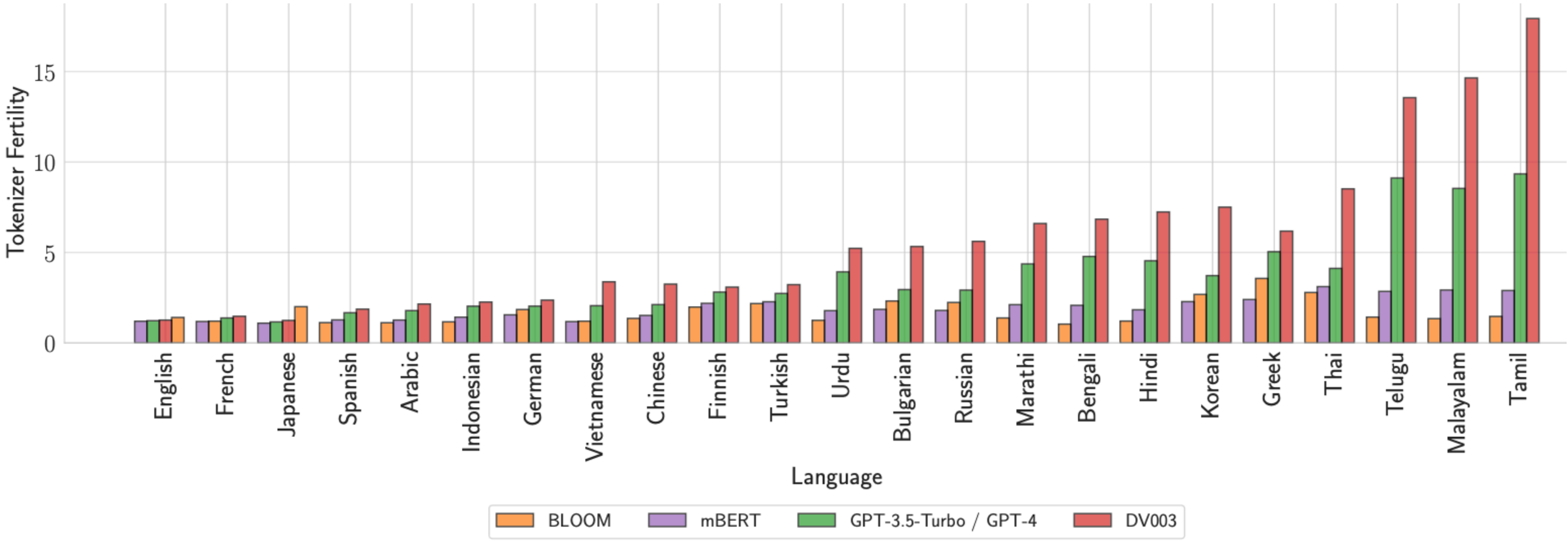
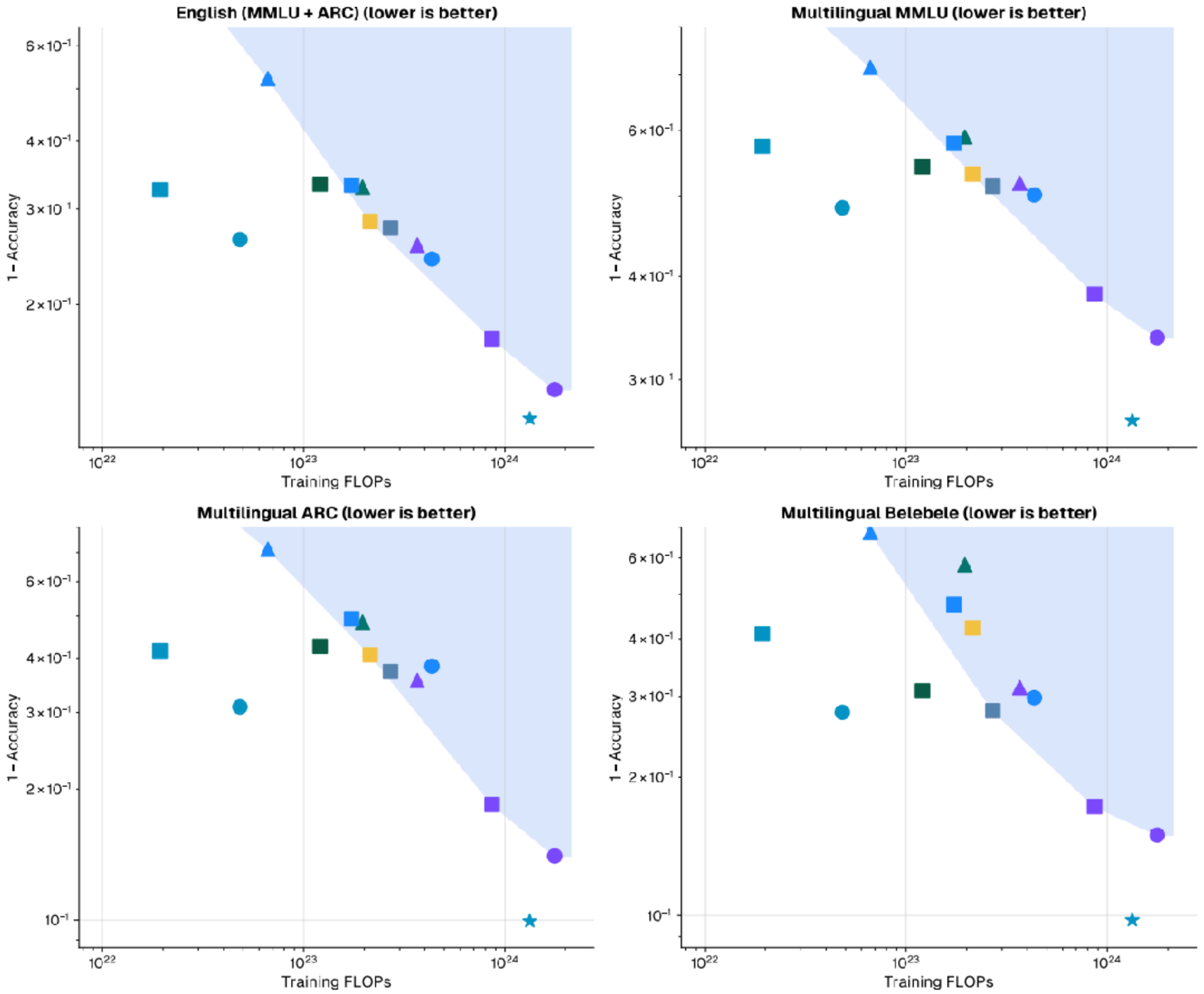


Image from "MEGA: Multilingual Evaluation of Generative AI", Ahuja et al., EMNLP 2023

Low Pretraining Diversity View is Incomplete: Role of Multilingual Data Curation



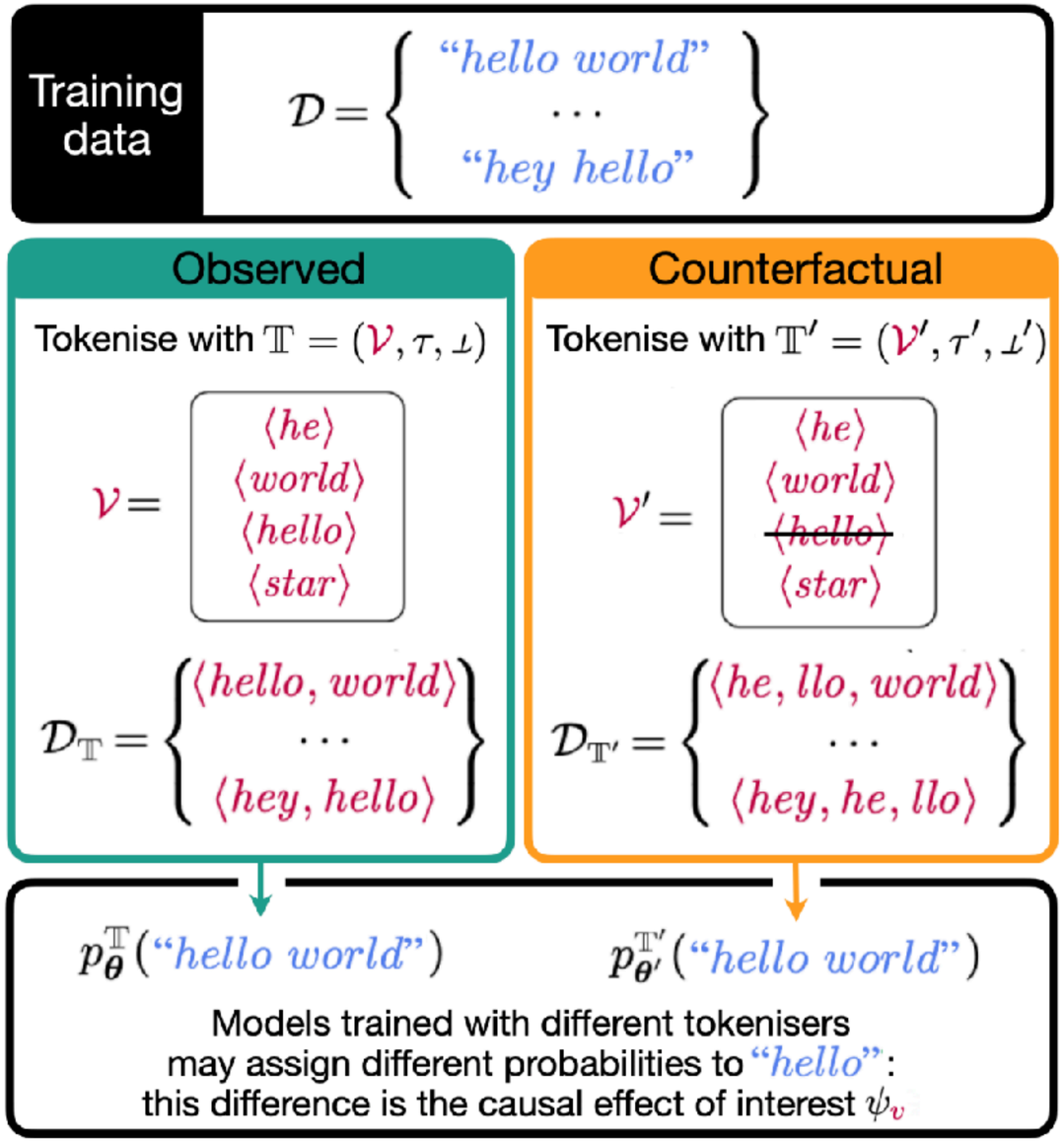
- New compute-performance Pareto frontier for multilingual capabilities
- Achieved via bespoke multilingual curation and high-quality translations
- Under 1T-token training budget, ≈8% allocated to multilingual data

[1] Figure from DatologyAI Team, "UberWeb: Insights from Multilingual Curation for a 20-Trillion-Token Dataset", https://arxiv.org/pdf/2602.15210v3, 2026

Tokenization in Multilingual LLMs

- Recall LLMs use *tokenizers* that deterministically map text into a sequence of *tokens*
  - *E.g.,* "A substantial improvement" → *A sub stantial im prove ment*
- Given a token vocabulary, there are many valid tokenizations of the same string
  - *E.g.,* "improvement" → */ im prove ment /, / improve ment /,* etc.
- However, LLMs use a single "canonical" tokenization
- LLMs share a single massively multilingual tokenizer with a fixed vocabulary
- The fixed vocabulary budget degrades tokenization quality in low-resource languages: Results in overfragmentation / higher token fertility (i.e., higher average number of tokens per word)
  - E.g., "सोमवार" → 'स' 'ोम' 'व' 'ार'

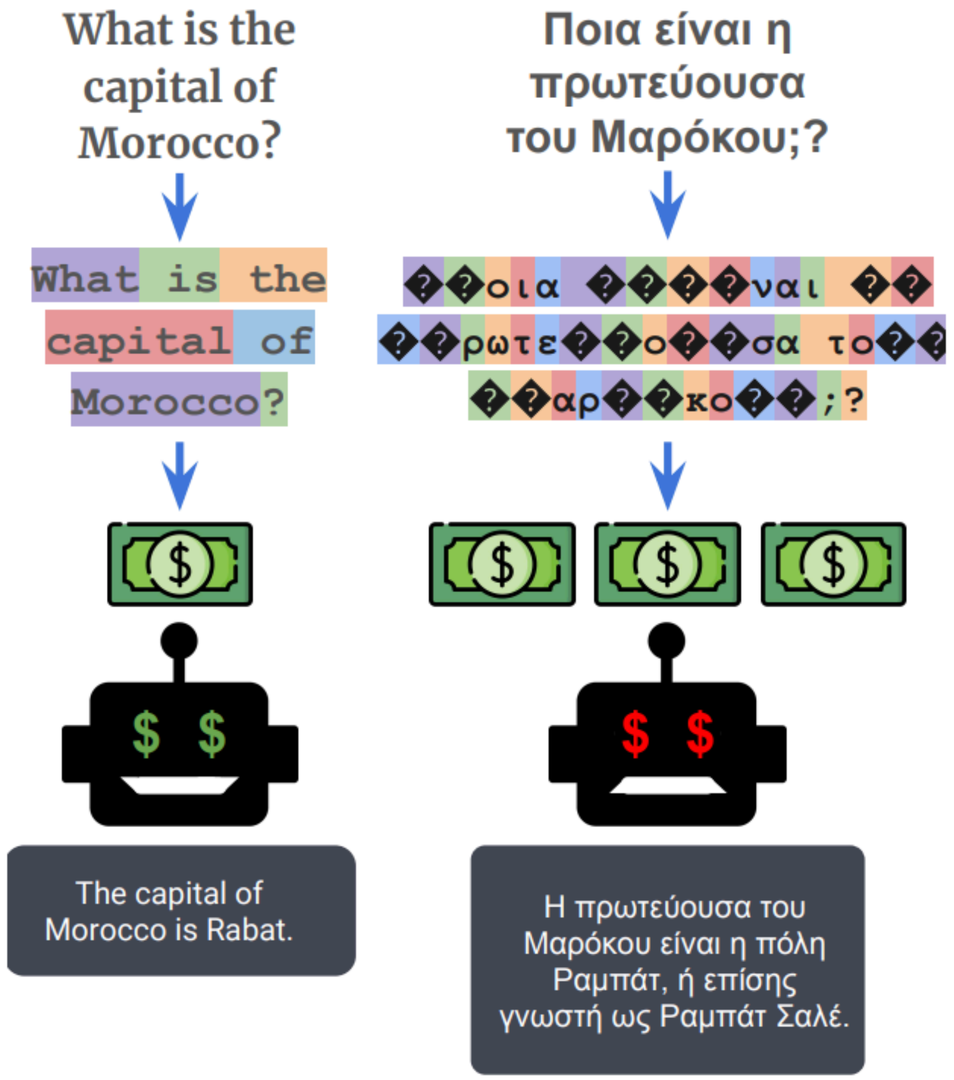
Why is High Token Fertility a Problem? (I)



- An LLM should ideally assign the same probability to a string regardless of tokenization
- In reality, a word's probability changes if its subword was removed from the token vocabulary. [1] characterizes this delta as "*tokenization bias*" [1]
- **Key finding**: Text represented as a single subword receives more log-probability than if split into two, and this bias grows during training

[1] Figure from Lesci et al., "Causal Estimation of Tokenisation Bias", ACL 2025

Why is High Token Fertility a Problem? (II)

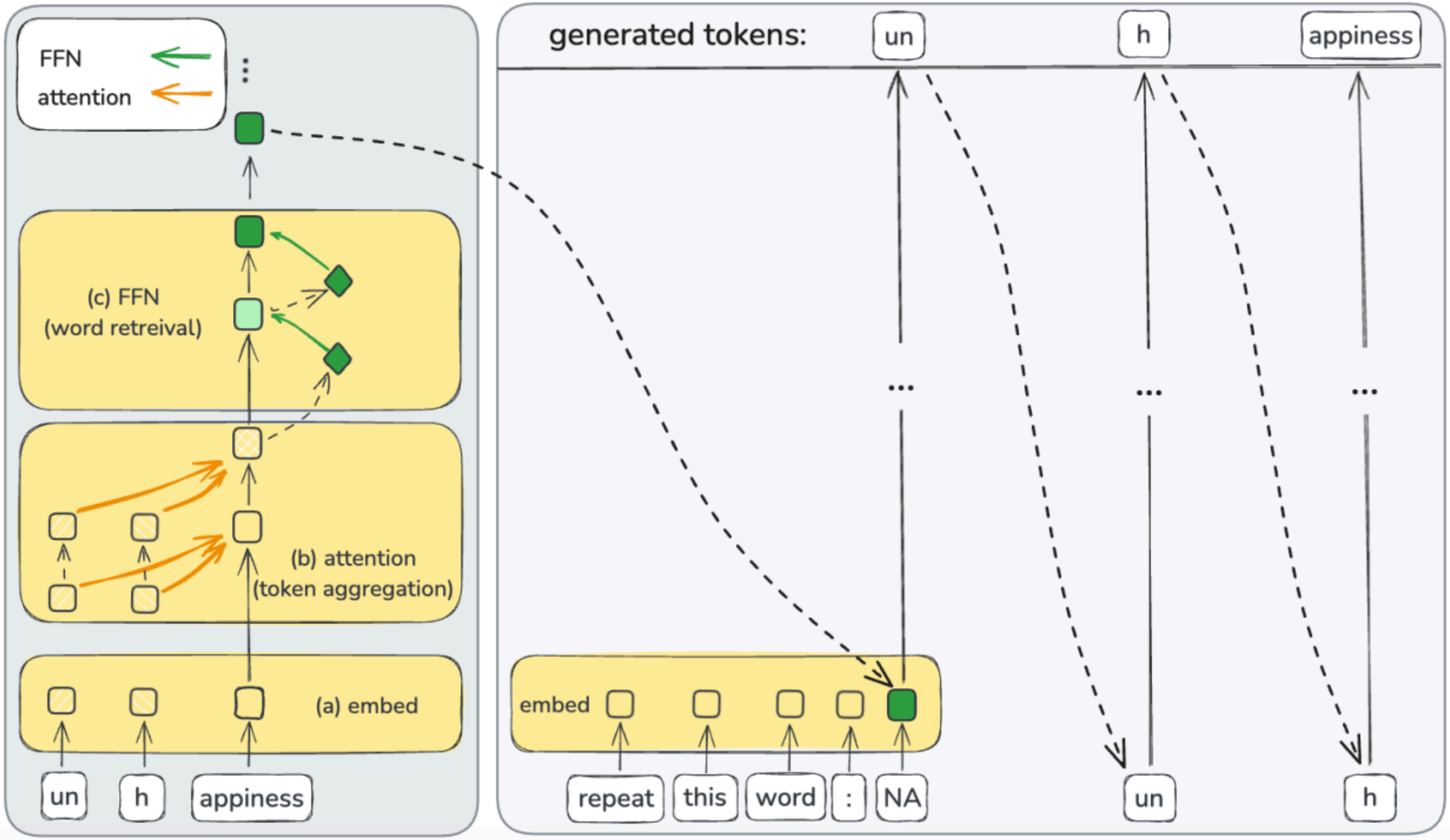


- LLM API providers charge by the token
- This directly increases cost of API usage for certain languages, even if the semantic content is the same

[1] Figure from Ahia et al., "Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models", EMNLP 2023

How do LLMs Internally Encode Words?

- ***Detokenization*** [1] -- Process of internally composing word representations from tokens



- Attention aggregates information from the word's previous tokens into the *last token*'s representations

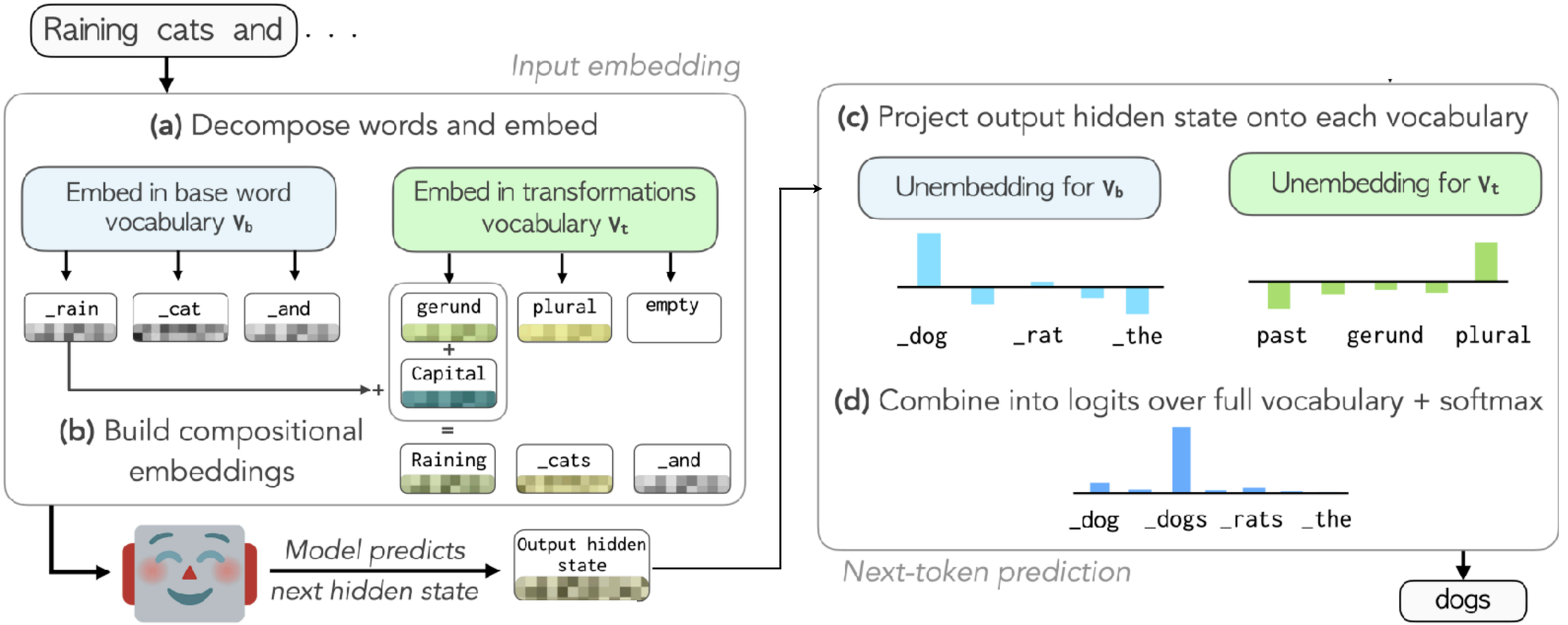
- FFN updates further promote the full word's concept in the last token's representation

[1] Elhage et al., "Softmax linear units", https://transformer-circuits.pub/2022/solu/index.html

Image from Kaplan et al., "From tokens to words: On the inner lexicon of LLMs", ICLR 2025

Compositional Vocabularies

- Token allocation is often wasteful with many tokens (e.g., *walks, walked, walking*), in high-resource languages like English, being surface variants of other words (e.g., *walk*)



- Structured vocabulary design: Each surface form is composed of base words and transformations [1]

[1] Reif et al., "Vocab Diet: Reshaping the Vocabulary of LLMs with Vector Arithmetic", https://arxiv.org/pdf/2510.17001

How do Multilingual LLMs Reason?

- Compute entropy of the full next token distribution when translating to Chinese
- Monitor the evolution of entropy and language probabilities (English, Chinese) across LLM layers
- **Phase I**: Building good token representations
- **Phase II**: Concept space with higher probability to tokens in English
- **Phase III**: Predict concepts in the chosen target language

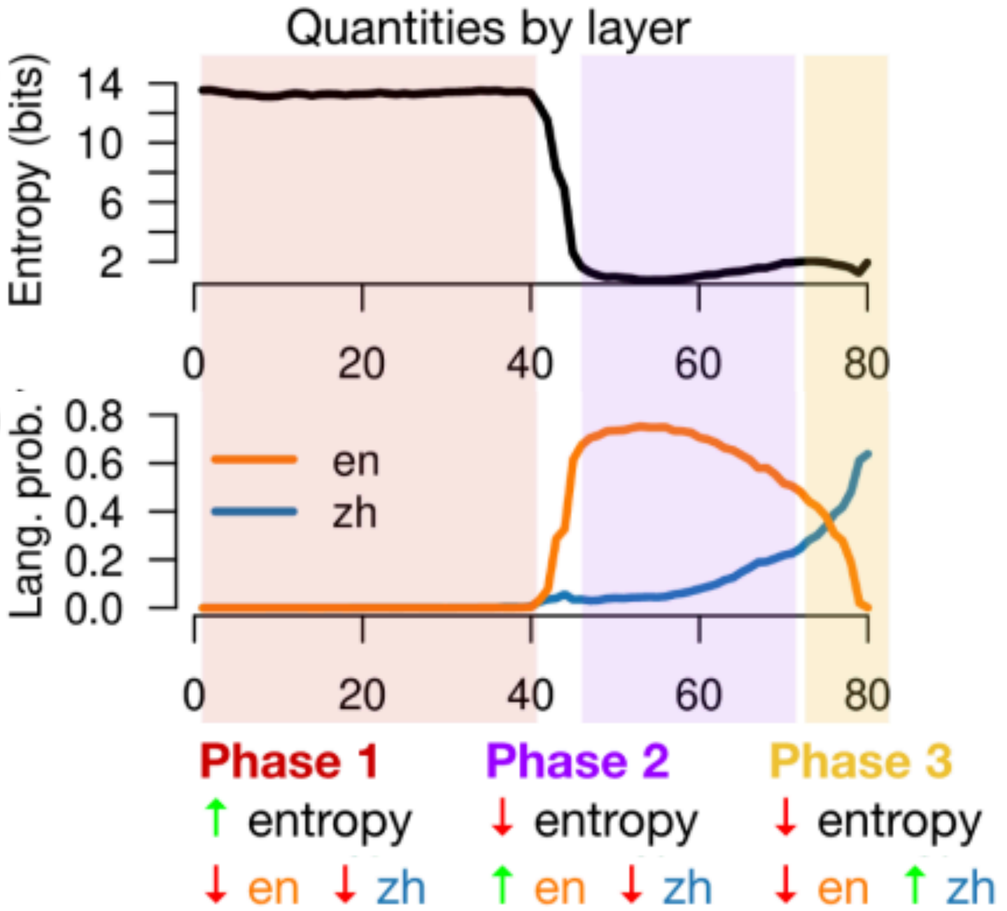
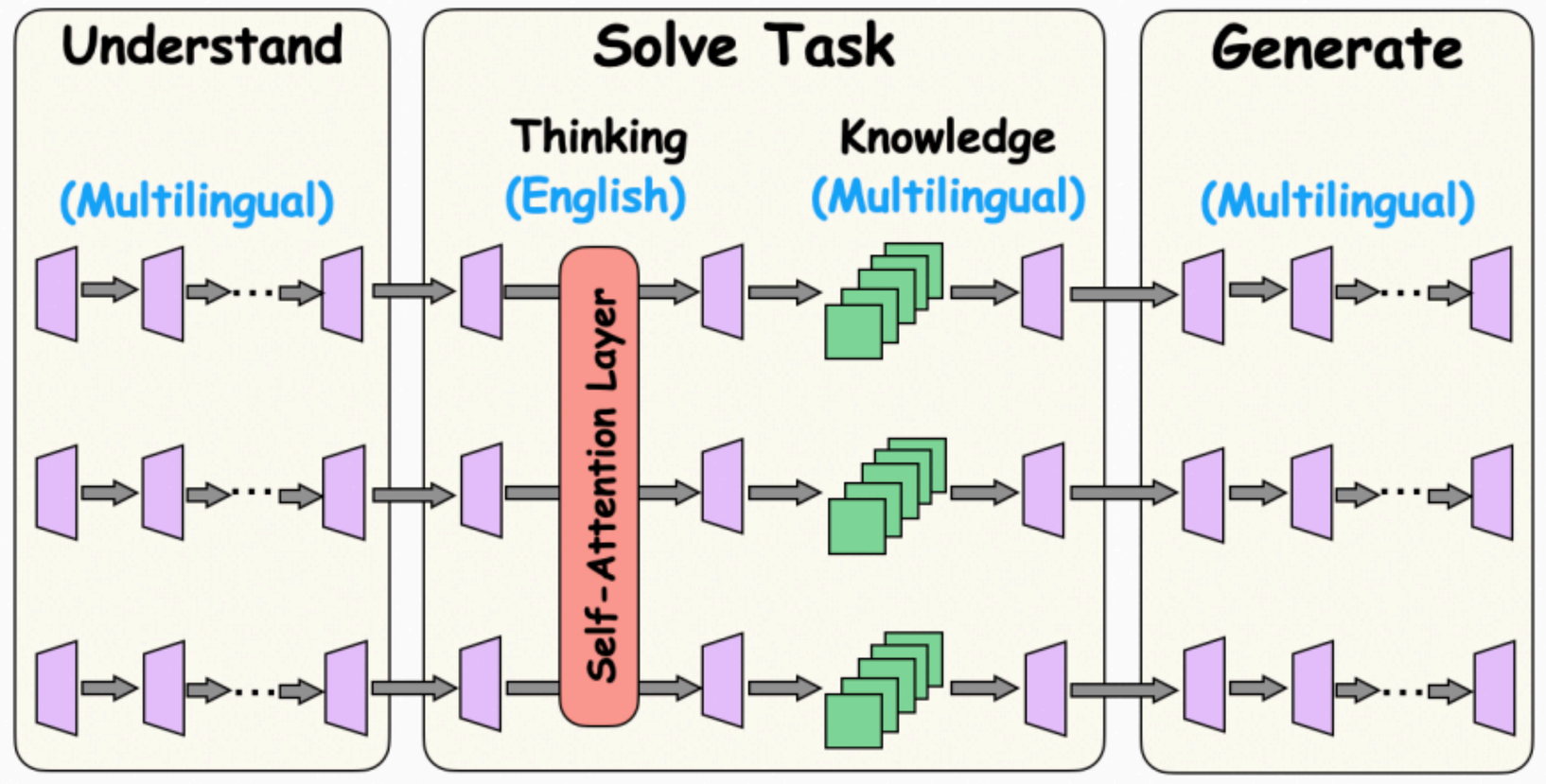


Image from Wendler et al., "Do Llamas Work in English? On the Latent Language of Multilingual Transformers", ACL 2024

Multilingual Workflow

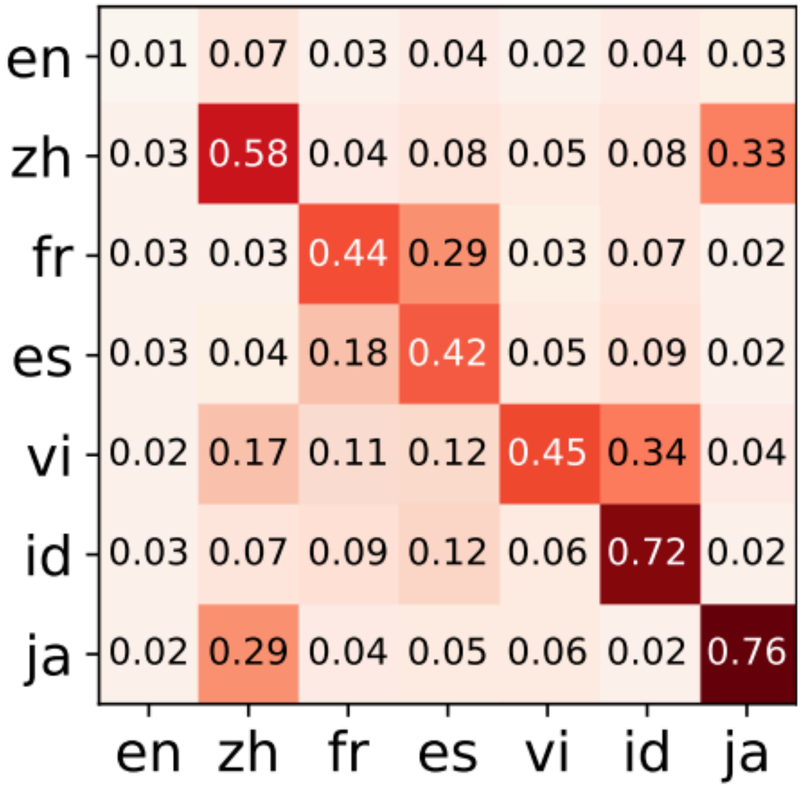


- Understand the non-English query, interpret in English, solve the task in English and generate outputs in the original language

Image from "How do Large Language Models Handle Multilingualism?", Zhao et al., NeurIPS 2024

Evidence for Language-specific Neurons?

- For the jth neuron in the ith layer, compute its *activation probability* by passing text in language k as input
- Get a probability distribution across languages and find its entropy (LAPE)
- Neurons with low LAPE scores are language-specific neurons



Perplexity change on deactivating the language-specific neurons (i.e., set to 0) of each row language

Question

¿Cómo puedo mejorar mis habilidades de gestión del tiempo?

(*Translation*: How can I improve my time management skills?)

Steered output in Chinese

Time管理是一个非常重要的技能，因为它可以提高你的生产力并使你更有效地完成任务。下面是一些提示来改进你的时间管理技能：1、制定计划：每天开始时制定清单并将任务分配给特定时间段。这样可以确保你在每天都有足够的时间来完成所有任务。2、设置目标：设置明确而可实现的目标可以帮助你保持专注并提高效果。3、抵制分心：当你正在工作时，请不要被其他事情分心。4、利用技术：……

(*Translation*: Time management is a very important skill because it increases your productivity and allows you to complete tasks more efficiently. Here are some tips to improve your time management skills: 1. Make a plan: Make a list at the beginning of each day and assign tasks to specific time periods. This ensures that you have enough time each day to complete all your tasks. 2. Set goals: Setting clear and achievable goals can help you stay focused and improve your results. 3. Resist distractions: When you are working, please don't be distracted by other things. 4. Utilize technology: ...)

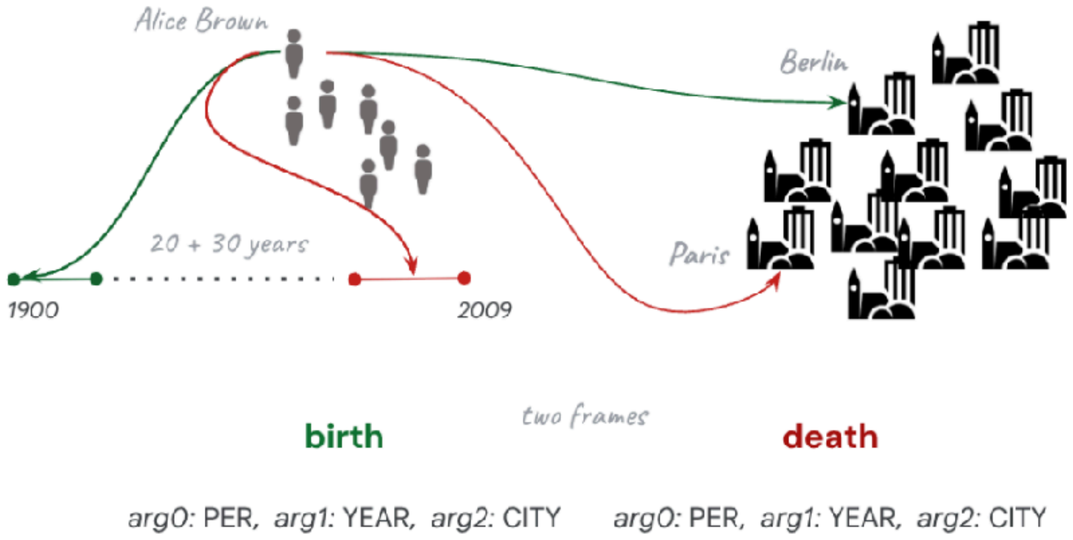
Q in Spanish: Activate Chinese, deactivate Spanish neurons

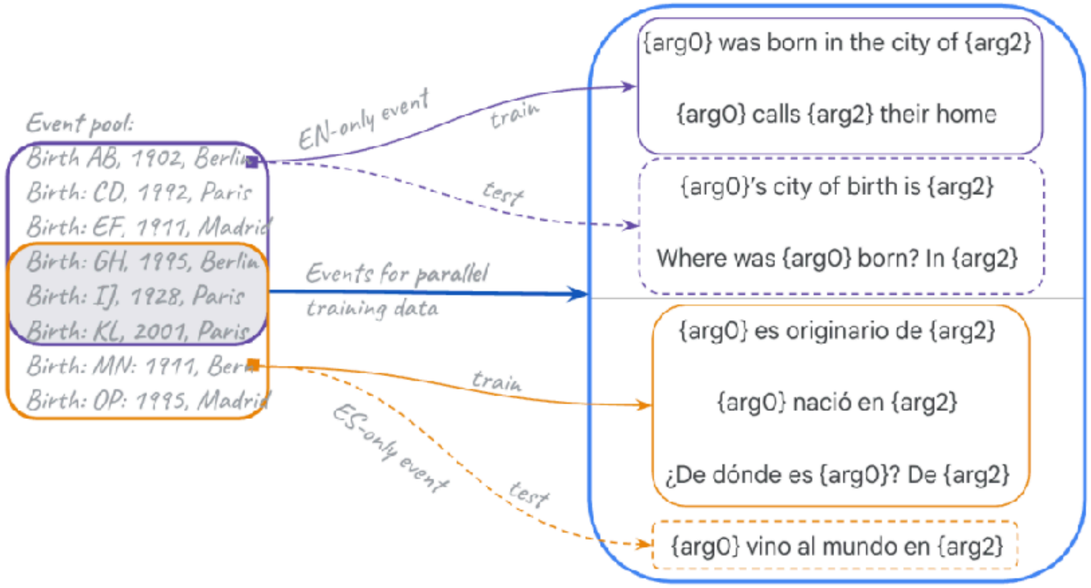
Images from "Language-Specific Neurons: The Key to Multilingual Capabilities in Large Language Models", Tang et al., ACL 2024

Cross-Lingual Transfer

- Cross-lingual transfer or cross-lingual generalisation is when data in high-resource languages is used to improve task performance in low-resource languages

- Studying cross-lingual factual recall in a "petri dish" [1]:





[1] Blum et al., "Beyond the Rosetta Stone: Unification Forces in Generalization Dynamics", https://arxiv.org/pdf/2508.11017, 2025

Cross-Lingual Transfer

- Cross-lingual transfer or cross-lingual generalisation is when data in high-resource languages is used to improve task performance in low-resource languages

- Studying cross-lingual factual recall in a petri dish [1]:
  - Cross-lingual factual recall fails when language identity is strongly encoded in the hidden representations. For example, texts written in Marathi will reference Marathi dishes much more than texts in English.
  - SoTA models can transfer knowledge between languages with a shared script, even if they're not related via language families (e.g., English and Indonesian).
  - To improve cross-lingual recall: Make differences between languages less obvious, balance attributes' frequencies across languages during pretraining
    - However, should not suppress the language feature unconditionally: Useful for queries that demand language- or culture-specific answers

[1] Blum et al., "Beyond the Rosetta Stone: Unification Forces in Generalization Dynamics", https://arxiv.org/pdf/2508.11017, 2025

Key Takeaways

- Effective multilingual support and multilingual data curation during pretraining can yield significant performance gains

- Tokenization poses many interesting challenges:

  - *Are LLMs robust to non-canonical tokenizations?* Turns out this is true for English [1].

  - *Transcend a fixed vocabulary?* Byte-based tokenization encodes bytes into dynamically-sized patches [2]

- Understanding cross-lingual transfer, the causes of cross-lingual hallucinations and designing mitigations are all open problems.

[1] Zheng et al., "Broken Tokens? Your Language Model can Secretly Handle Non-Canonical Tokenizations", NeurIPS 2025
[2] Pagnoni et al., "Byte Latent Transformer: Patches Scale Better Than Tokens", ACL 2025

Speech and LLMs

Spoken Language Models

- Spoken LMs: Integrate speech and LLMs to support understanding and generation tasks
  - Also referred to as large audio LMs, speech LLMs, multimodal LLMs, etc.

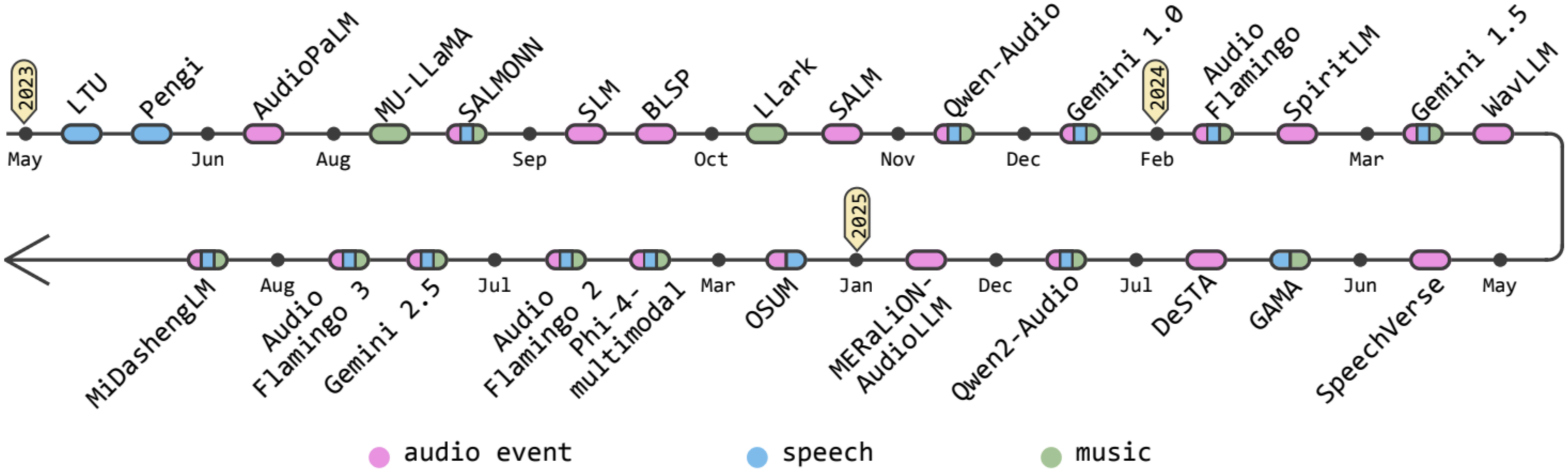
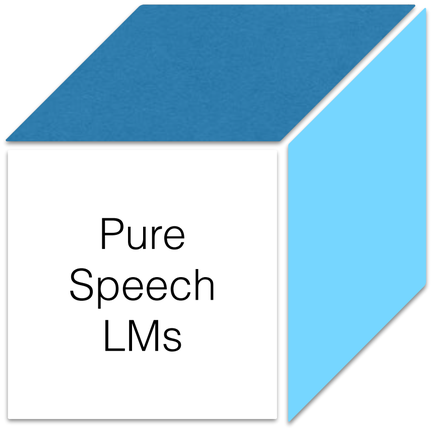


Image from "Towards General Auditory Intelligence: LLMs for Machine Listening and Speaking", Wang et al., https://arxiv.org/pdf/2511.01299, Nov 2025

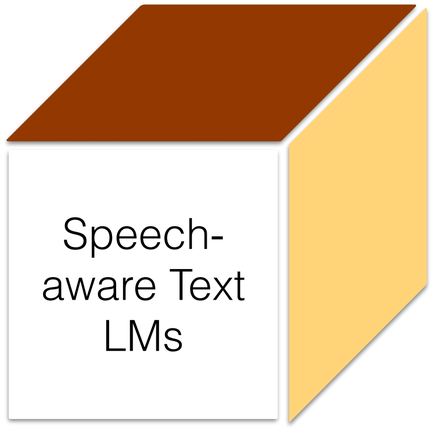
Typology of Spoken LMs [1]



- Trained on unlabelled speech data; no text involved
- Trained using a next-token objective function
- Examples: AudioLM, TWIST, etc.



- Trained jointly on speech and text
- Trained using paired speech-text data, unlabelled speech and text
- Examples: Moshi, Mini-Omni, etc.

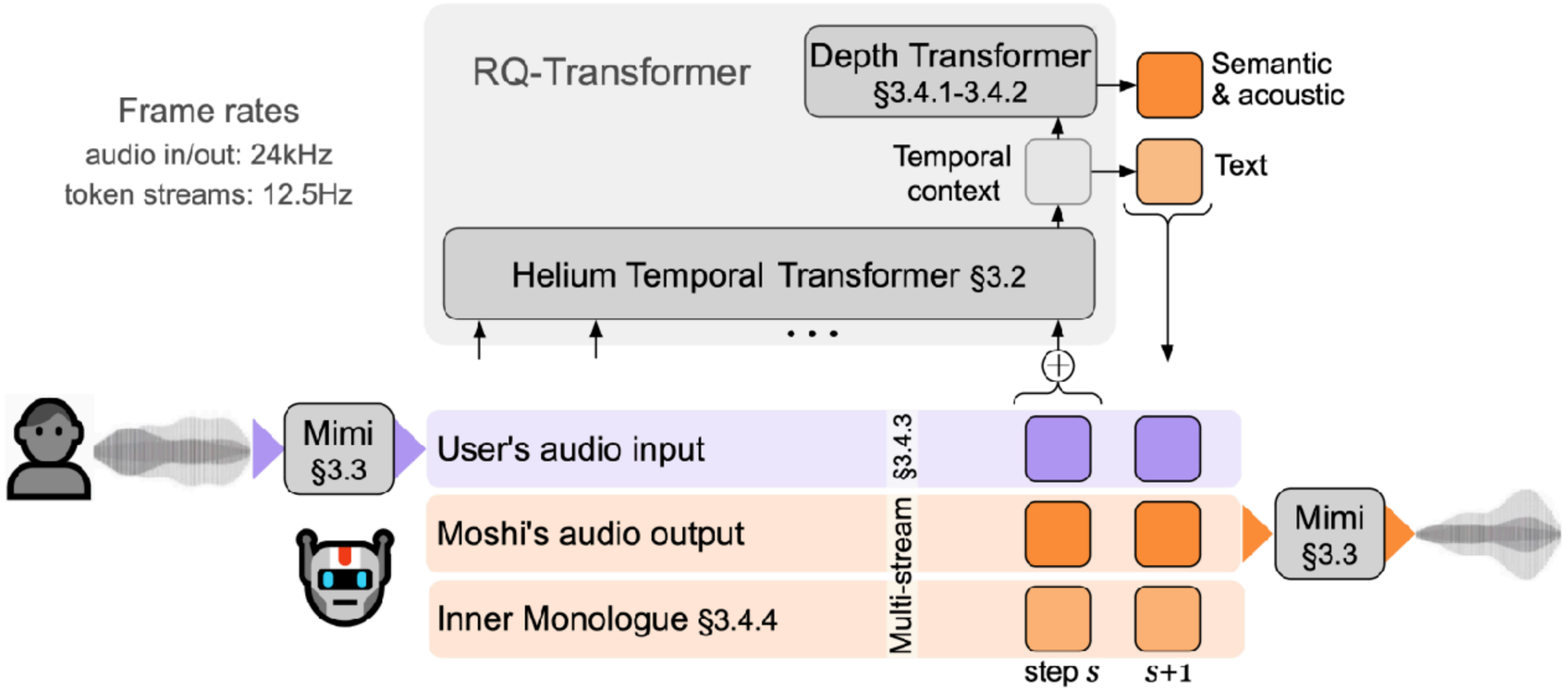


- Combine text LLMs with pretrained speech encoders
- Trained using paired speech-text data
- Examples: SALMONN, SALSA, Qwen-Audio-Chat, etc.

[1] Arora et al., "On The Landscape of Spoken Language Models: A Comprehensive Survey", TMLR 2025

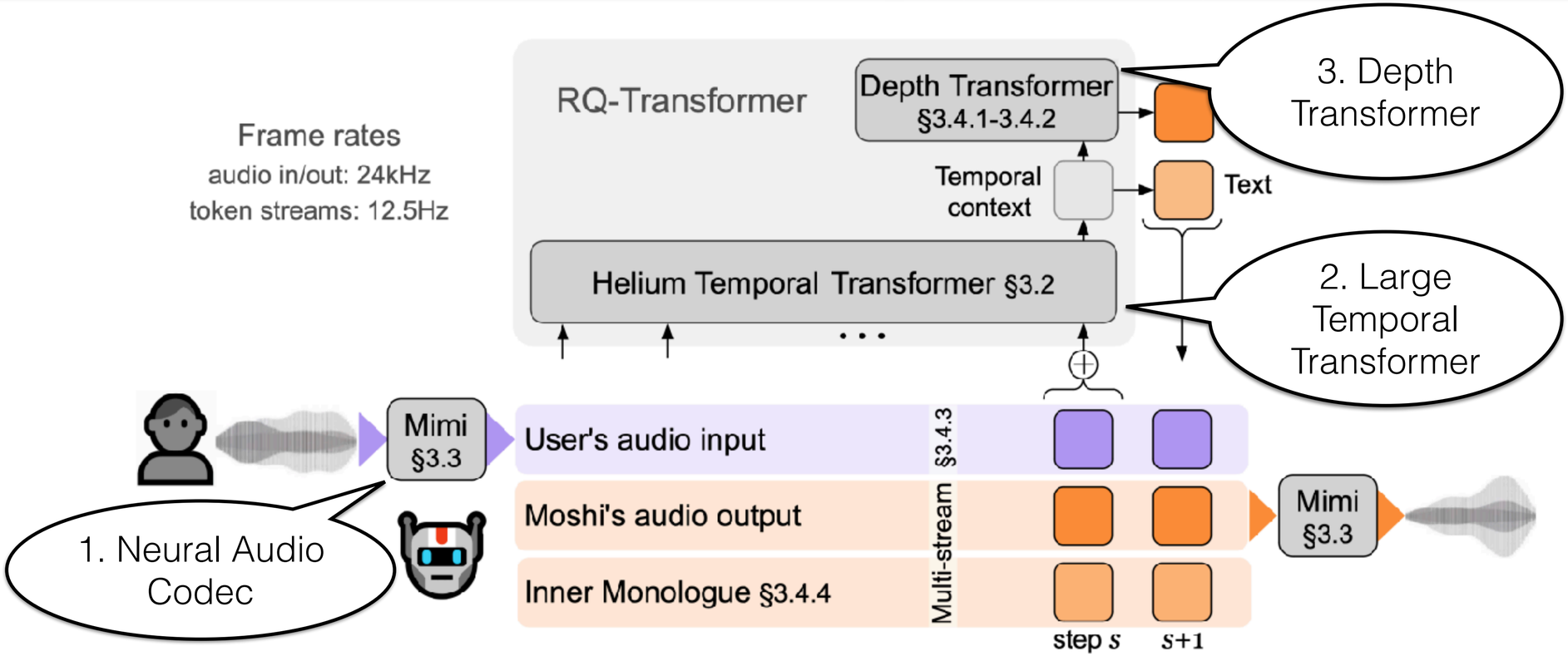
Moshi

- Moshi is an end-to-end speech-to-speech foundation model that enables real-time spoken dialogues



[1] Défossez et al., " Moshi: a speech-text foundation model for real-time dialogue", https://arxiv.org/pdf/2410.00037, 2024
github.com/kyutai-labs/moshi

Moshi

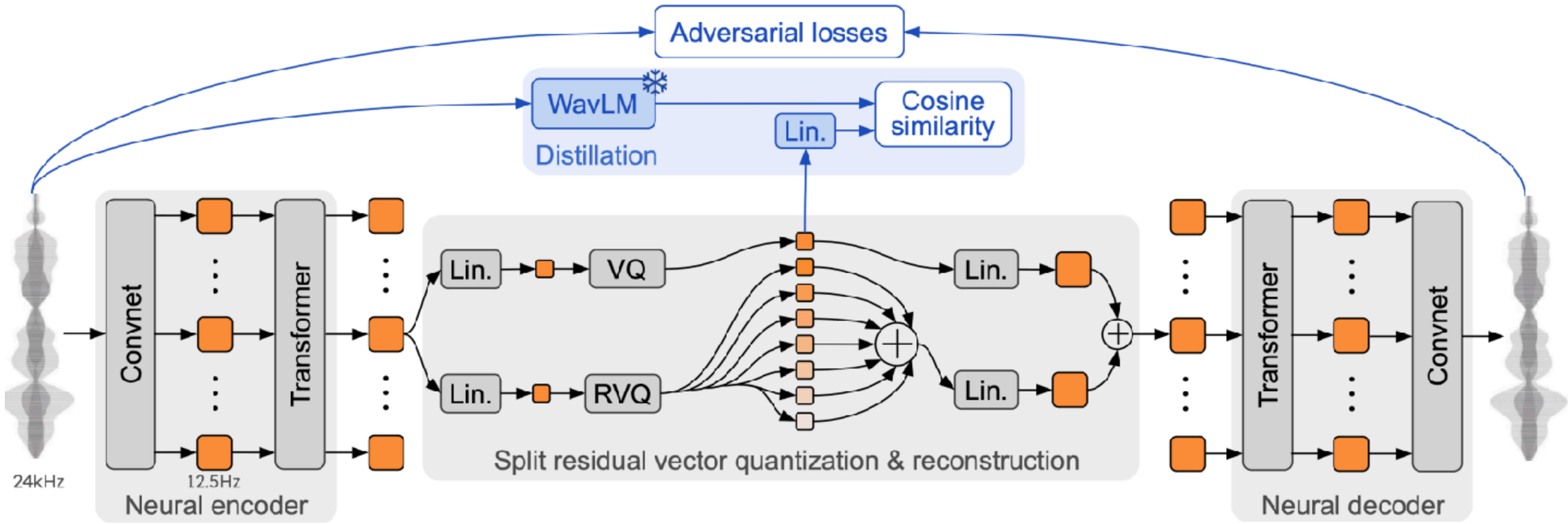


- Main components of Moshi:
  1. A *neural audio codec* used to tokenize speech
  2. Large temporal transformer initialized from a classic text LLM (Helium-7B)
  3. Depth transformer that generates multi-stream tokens within a time-step

  } RQ-Transformer

[1] Défossez et al., " Moshi: a speech-text foundation model for real-time dialogue", https://arxiv.org/pdf/2410.00037, 2024

Moshi: Neural Audio Codec



- Encoder: Converts raw audio into latent vectors
- Residual Vector Quantisation (RVQ): Approximate the encoder latents using vector quantization in multiple rounds, where each round acts on the residual from the previous round
- Decoder: Reconstructs audio from tokens

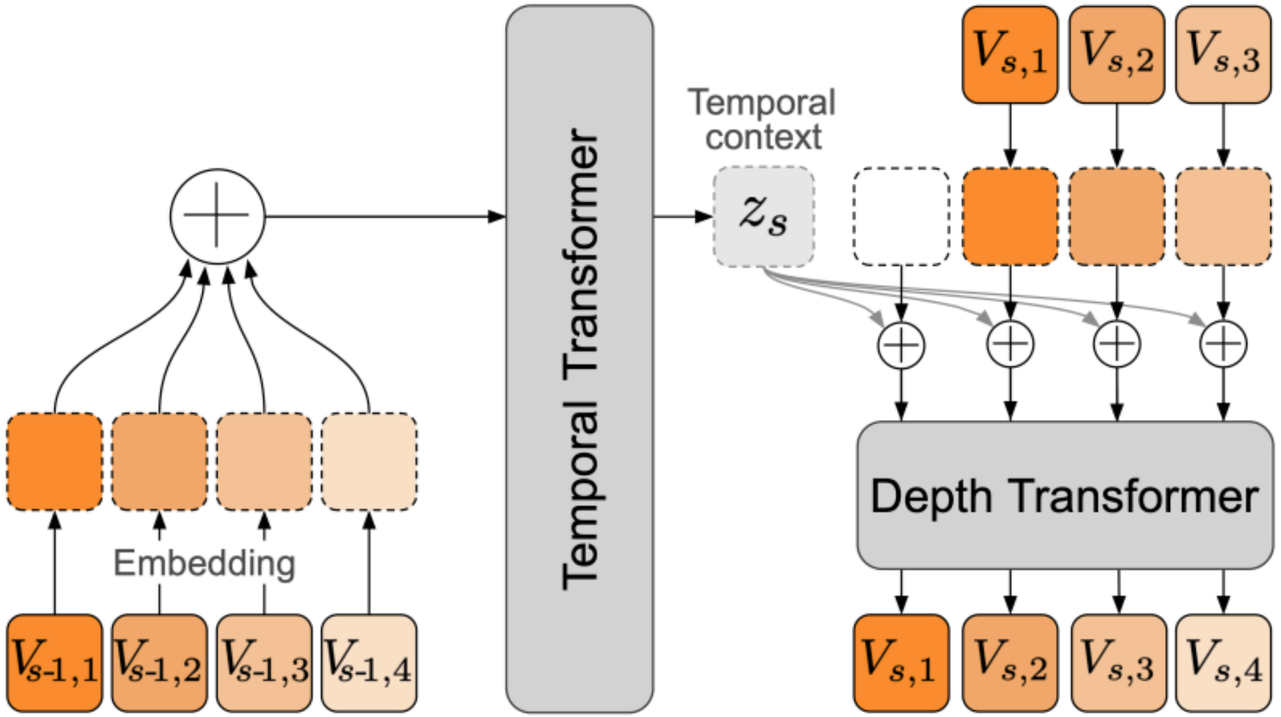
Algorithm 1: Residual Vector Quantization

Input: $y = \text{enc}(x)$ the output of the encoder, vector quantizers Q_i for $i = 1..N_q$
Output: the quantized $\hat{y}$
$\hat{y} \leftarrow 0.0$
residual $\leftarrow y$
for $i = 1$ *to* N_q **do**
  $\hat{y}$ += Q_i(residual)
  residual −= Q_i(residual)
return $\hat{y}$

Image from Défossez et al., " Moshi: a speech-text foundation model for real-time dialogue", https://arxiv.org/pdf/2410.00037, 2024

[2] Algorithm from Zeghidour et al., " SoundStream: An End-to-End Neural Audio Codec", IEEE Transactions on Audio,Speech and Language Processing, 2021

Moshi: RQ-Transformer



- Two-level autoregressive factorization
- At each time-step indexed by s, the temporal transformer produces a contextual embedding z_s and an overall sequence of length S
- A smaller depth transformer is conditioned on z_s over K steps
- This design allows Moshi to handle multi-stream tokens

Image from Défossez et al., " Moshi: a speech-text foundation model for real-time dialogue", https://arxiv.org/pdf/2410.00037, 2024

Interesting Challenges with Spoken LMs

- Insufficient acoustic grounding: Leads to "modality gap" between speech and text [1]
- Discourse-level reasoning with correct understanding of user intent needs to be improved; also need more *full-duplex* (i.e. speaking and listening simultaneously) benchmarks to evaluate this capability. [2]
- A lot more variability in speech: Accents, Dialects, Environment noise, Microphone quality. Existing spoken LMs are not fully invariant to these factors yet.
- Real-time voice interactions require low latency. Thinking, after listening to the entire query, increases latency. Active area of research to enable simultaneous hearing and thinking in spoken LMs. [3]

[1] Wang et al., "Closing the Modality Reasoning Gap for Speech Large Language Models", https://arxiv.org/abs/2601.05543, Accepted to ACL 2026
[2] Ray et al., "τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains", https://arxiv.org/abs/2603.13686, 2026
[3] Chiang et al., "SHANKS: Simultaneous Hearing and Thinking for Spoken Language Models", https://arxiv.org/abs/2510.06917, Accepted to ACL 2026